Counterintuitive Result and How We Resolved It (Run 1 vs Run 2)

This document explains the initial “contradiction” between empirical Competing Risks plots and Cox model partial effects, and then summarizes what we did in Run 1 and Run 2 to drill down and identify the underlying drivers.

Note on interpretation: Cox model coefficients and interaction effects are **hazard (instantaneous risk) effects**, not direct changes in promotion probability.

The Initial Contradiction

In the empirical Competing Risks (CIF) plots, we observe that `tb_ratio_rank_norm_snr` appears **positively** related to promotion (higher values show higher promotion incidence).
But in the Cox model partial effects (CELL 12.6), the relationship appears to **reverse** when `cum_tb_rcvd_ratio_snr` is in the model.

That looks contradictory at first glance.

Why the Contradiction Happens

The two plots answer **different questions**:

- **Competing Risks (CIF)** plots show **raw, unadjusted** patterns by group.
- **Cox partial effects** show **conditional, adjusted** effects: the model asks “what happens if we change one variable while holding the others fixed?”

Because `tb_ratio_rank_norm_snr` and `cum_tb_rcvd_ratio_snr` are correlated, the CIF plot reflects their **combined advantage**, while the Cox model separates them and assigns shared signal to one or the other. This is a standard confounding/collinearity situation.

Run 1: Cumulative TB vs Pooled TB

Goal: Compare cumulative TB exposure to pooled mean TB while allowing interaction.

Variables: - `z_cum_tb_rcvd_ratio_snr` - `z_pool_tb_ratio_mean_snr` - `z_star_pool_interaction`

What this isolates:
Whether cumulative TB or pooled TB is the dominant signal once both are modeled together, and how their effects trade off at constant hazard.

Interaction variable explanation (`z_star_pool_interaction`):
This term captures whether the effect of cumulative TB depends on pooled TB (and vice versa). In practice, it tests whether the slope of `z_cum_tb_rcvd_ratio_snr` changes across levels of `z_pool_tb_ratio_mean_snr`. If the interaction coefficient is: - **Positive**, the

effect of one variable becomes stronger as the other increases. - **Negative**, the effect of one variable becomes weaker (or can even flip) as the other increases.

In the Cox model:

$$\log(HR) = b_1 x + b_2 y + b_3 xy$$

with $x = z_cum_tb_rcvd_ratio_snr$ and $y = z_pool_tb_ratio_mean_snr$, the interaction term b_3 tells us whether cumulative TB and pooled TB **reinforce** each other (synergy) or **substitute** for each other (trade-off).

Run 2: Rank-Normalized TB vs Pooled TB

Goal: Replace cumulative TB with rank-normalized TB and see whether the rank-normalized signal holds once pooled TB is included.

Variables: - `z_tb_ratio_rank_norm_snr` - `z_pool_tb_ratio_mean_snr` - `z_rank_pool_interaction`

What this isolates:
Whether the rank-normalized TB metric adds independent explanatory power, or whether the positive empirical association is mostly explained by pooled TB.

Interaction variable explanation (`z_rank_pool_interaction`):
This term tests whether the effect of rank-normalized TB depends on pooled TB. It answers: "Does rank-based TB matter more (or less) for promotion when pooled TB is high?" If the interaction coefficient is: - **Positive**, the rank-normalized TB effect strengthens at higher pooled TB. - **Negative**, the rank-normalized TB effect weakens at higher pooled TB.

In the Cox model:

$$\log(HR) = b_1 x + b_2 y + b_3 xy$$

with $x = z_tb_ratio_rank_norm_snr$ and $y = z_pool_tb_ratio_mean_snr$, the interaction term b_3 quantifies whether rank-normalized TB and pooled TB act **multiplicatively** (interaction present) rather than **additively** (no interaction).

What We Found

- The empirical (CIF) plots show the **net, real-world association**, which can be positive because TB metrics move together.
- The Cox model shows the **conditional effect**, which can **flip** once the correlated TB metric is held fixed.
- Run 1 and Run 2 clarify **which TB construct carries the dominant signal**, and whether the alternative metric is redundant or reverses direction once adjusted.
- The interaction terms allow us to quantify **how much change in one metric offsets the other** at constant hazard:
  - If the fitted model is

    $$\log(HR) = b_1 x + b_2 y + b_3 xy$$

    then:

    $$\frac{\partial \log(HR)}{\partial x} = b_1 + b_3 y, \quad \frac{\partial \log(HR)}{\partial y} = b_2 + b_3 x$$

- If $b_3 = 0$, the trade-off at constant HR simplifies to:

$$\frac{dx}{dy} = -\frac{b_2}{b_1}$$

Specific Results From the Slides

The slides include concrete coefficient values. Below, those specific results are interpreted in plain language.

Run 1 (cumulative TB vs pooled TB)

From the slide coefficient table (interaction included): - $b_1 = 0.2413$ for `z_cum_tb_rcvd_ratio_snr` - $b_2 = 0.3125$ for `z_pool_tb_ratio_mean_snr` - $b_3 = -0.0493$ for `z_star_pool_interaction`

Interpretation (z-score units): - Both main effects are positive, but the **interaction is negative**. This means the joint effect is **less than additive**: when both cumulative TB and pooled TB are high, the combined boost to hazard is dampened. - The marginal effect of cumulative TB is:

$$\frac{\partial \log(HR)}{\partial x} = 0.2413 - 0.0493 \cdot y$$

so the benefit of cumulative TB **shrinks as pooled TB increases**. - The marginal effect of pooled TB is:

$$\frac{\partial \log(HR)}{\partial y} = 0.3125 - 0.0493 \cdot x$$

so the benefit of pooled TB **shrinks as cumulative TB increases**.

Effect-size illustration (interaction strength):
If pooled TB moves from **–1 SD to +1 SD**, the marginal effect of cumulative TB changes by:

$$2 \cdot b_3 = 2 \cdot (-0.0493) = -0.0986$$

That corresponds to a **~9.4% decrease** in the hazard multiplier for a 1-SD change in cumulative TB across that range (since $e^{-0.0986} \approx 0.906$).
This is a **hazard (instantaneous risk) interpretation**, not a direct change in **promotion probability**.

Effect sentence (HR audience):
The hazard multiplier (a short-term promotion rate, not a long-run probability) for **individual cumulative TB** is **smaller** when the senior-rater pool is stronger. In numbers: a **1-SD increase in cumulative TB** multiplies hazard by about **1.34×** when pooled TB is **–1 SD** ($e^{0.2413-0.0493\cdot(-1)} \approx 1.34$), but only about **1.21×** when pooled TB is **+1 SD** ($e^{0.2413-0.0493\cdot(1)} \approx 1.21$) — a **~9.4% reduction** in the hazard multiplier across that range. In other words, **being a high performer in a strong pool still helps**, but the boost is **dampened** compared with being a high performer in a weaker pool.

Run 2 (rank-normalized TB vs pooled TB)

From the slide coefficient table (interaction included): - $b_1 = 0.2988$ for `z_tb_ratio_rank_norm_snr` - $b_2 = 0.4073$ for `z_pool_tb_ratio_mean_snr` - $b_3 = 0.0087$ for `z_rank_pool_interaction`

Interpretation (z-score units): - Both main effects are positive, and the **interaction is slightly positive**. This implies mild **synergy**: when both rank-normalized TB and pooled TB are high, the combined effect is a bit stronger than additive. - The

marginal effect of rank-normalized TB is:

$$\frac{\partial \log(HR)}{\partial x} = 0.2988 + 0.0087 \cdot y$$

so the rank-normalized TB effect **strengthens slightly** as pooled TB increases. - The marginal effect of pooled TB is:

$$\frac{\partial \log(HR)}{\partial y} = 0.4073 + 0.0087 \cdot x$$

so the pooled TB effect **strengthens slightly** as rank-normalized TB increases.

Effect-size illustration (interaction strength):
If pooled TB moves from **–1 SD to +1 SD**, the marginal effect of rank-normalized TB changes by:

$$2 \cdot b_3 = 2 \cdot 0.0087 = 0.0174$$

That corresponds to a **~1.8% increase** in the hazard multiplier for a 1-SD change in rank-normalized TB across that range (since $e^{0.0174} \approx 1.018$).
This is a **hazard (instantaneous risk) interpretation**, not a direct change in **promotion probability**.

Effect sentence (HR audience):
The hazard multiplier (a short-term promotion rate, not a long-run probability) for **rank-based TB** is **slightly larger** when the senior-rater pool is stronger. In numbers: a **1-SD increase in rank-based TB** multiplies hazard by about **1.34×** when pooled TB is **–1 SD** ($e^{0.2988+0.0087\cdot(-1)} \approx 1.34$), and about **1.36×** when pooled TB is **+1 SD** ($e^{0.2988+0.0087\cdot(1)} \approx 1.36$). So **high performers in strong pools are not penalized**; if anything, the model shows a **small additional boost** when pool strength is higher.

Counterintuitive reversal (rank-normalized TB)

The slides highlight a reversal for `tb_ratio_rank_norm_snr`: - **Empirical CIF:** higher rank-normalized TB appears to correlate with higher promotion incidence. - **Cox partial effects:** once cumulative TB (or pooled TB) is included, the conditional effect of rank-normalized TB becomes weaker or reverses.

Interpretation:
The raw CIF plot reflects a **combined advantage** because the TB constructs move together. The model isolates the **independent contribution**, which can flip once the correlated TB metric is held fixed.

Takeaway for Interpretation

The “contradiction” is not an error; it is the difference between **raw association** and **conditional effect**.
Run 1 and Run 2 are the deliberate drill-down: they test which TB construct is really driving promotion outcomes, and they quantify trade-offs so we can interpret the relative magnitudes in z-score units.